

Phase A — primaries

ONE fused @compute dispatch

1 thread / primary electron

full history in a for-loop:

Born ion. · Emfietzoglou exc.

elastic · vibrational

no per-step dispatch (fused)

Phase B — secondaries

wavefront stepper

1 dispatch / physics step

advances all live secondaries

*cannot fuse: count unknown
until A completes*

Phase C — chemistry

Karamitros 9-reaction IRT

Web Worker (off main thread)

6 species · Smoluchowski +

Onsager-screened reactions

CPU-bound (the bottleneck)

shared GPU buffers: rad_buf (energy deposits) · secondary buffer

1283 voxel grid · fixed-point integer atomicAdd (×100/eV)

radical list → G-values, SSB/DSB